

Sentinel-DE: Automated Triage & Sync

The Hook: "Good morning/afternoon. Today, I'm excited to walk you through **Sentinel-DE**—a system I designed to solve a classic bottleneck in data engineering: the 'operational tax' of manual triage and incident reporting."

The Problem: "As our data infrastructure grows, so does the administrative noise. We've found ourselves spending valuable engineering hours acting as 'Air Traffic Controllers'—manually routing requests, chasing down asset owners, and prep-work for sync meetings."

The Solution: "Sentinel-DE is an agentic automation framework built to eliminate that friction. By leveraging grounded AI logic, it takes us from a high-touch manual process to an **exception-based handling model**."

The Roadmap: "Over the next few minutes, I'll show you how this system recovers over **one hour of capacity every single day**, the multi-pillar framework I used to evaluate its security and performance, and the measurable business value it is already delivering to the team."

Automating Technical Triage: High-Efficiency Incident Response

◆ The problem

As a Lead/senior engineer one of the daily tasks is to address the operational bottleneck of manual incident triage within a Data Engineering team. Specifically, four critical manual workflows listed below take place daily:

1. **Identifying system owners** for failing assets.
2. **Calculating business criticality** based on stakeholder impact.
3. **Routing alerts** to the correct communication channels.
4. **Synthesizing data for the Weekly Sync** by automatically filtering and logging "High" and "Critical" incidents into a centralized Google Sheet.

The Reality of the Role: "In a lead or senior engineering role, our most valuable asset is 'Deep Work' time. However, that time is constantly fragmented by what I call the 'Operational Tax'—the manual logistics of managing incident flow across a complex data stack."

The Four Friction Points: "Every day, we face four specific bottlenecks that drain our collective bandwidth:

Identity Search: Hunting through outdated documentation to find which engineer owns which failing asset.

Subjective Analysis: Manually calculating business criticality, which is often inconsistent depending on who is doing the triage.

Communication Overhead: The repetitive task of routing these alerts to the correct Slack or Teams channels.

Administrative Debt: The manual effort required to filter and log these incidents into our Weekly Sync Google Sheet."

The Cumulative Impact: "These aren't just small tasks; they are context-switching triggers. When we have to stop building a pipeline to play 'Air Traffic Controller,' we lose the momentum needed for our core engineering objectives. This project was designed to reclaim that lost time."

The Goal: "The objective is simple: move the team away from manual 'triage

logistics' and toward a system where we only focus on **High** and **Critical** events that have been pre-processed and pre-logged by Sentinel-DE."

Continuation

Why it matters

While these tasks are individually small, they collectively created a "productivity tax" that reduced the senior engineer's capacity for high-value engineering such as DE improvements, research, and pipeline optimization.

The "Capacity Recovery" Angle: "This project isn't just about 'saving time'—it's about **Capacity Recovery**. Every minute an engineer spends playing 'Air Traffic Control' in Slack or copy-pasting incident details into a spreadsheet is a minute they aren't spent on the high-value data modeling or SQL optimization that moves our business forward."

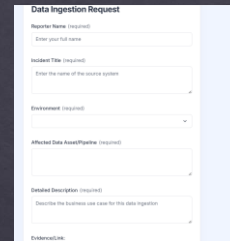
The Weekly Sync Impact: "A key feature of this automation is the 'Strategic Filter.' By automatically logging only High and Critical issues to our Google Sheet, we've essentially automated the meeting agenda. We no longer waste time identifying *what* to talk about; we spend the entire meeting discussing *how* to solve it."

The Bottom Line: "Ultimately, we've built an infrastructure that manages itself. We've turned a messy, manual process into a clean, auditable data stream, allowing our most expensive assets—our engineers—to stay focused on deep-work tasks."

Workflow Specification: The Lifecycle of an Incident: From Submission to Strategic Archive

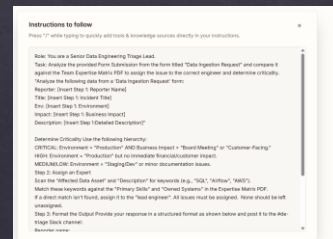
1. The Trigger: Structured Data Ingestion

- ◆ **Source** - "Data Ingestion Request" Form.
- ◆ **Mechanism** - Zapier captures structured metadata (Environment, Affected Asset, Business impact)



2. The Logic: AI-Agentic Reasoning

- ◆ **Step** - Sentinel-DE parses the submission against the **Team Expertise Matrix (PDF)**.
- ◆ **Action** - The Agent assigns a **Criticality Score** and identifies the specific **Primary Engineer** handle.



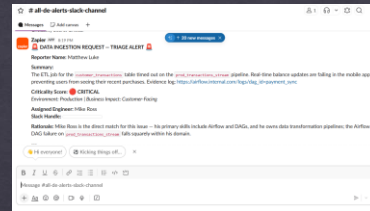
Phase 1 (The Trigger): "It all starts with structured input. By using a form instead of raw Slack messages, we ensure we have the clean data—like 'Environment' and 'Data Asset'—needed for high-accuracy AI processing."

Phase 2 (The Intelligence): "This is the 'brain' of the operation. The Agent doesn't just guess; it performs a vector search against our Expertise Matrix. It identifies that a failure in the prod_transactions_stream belongs to the Infrastructure team, specifically tagging the engineer responsible for that pipeline."

continuation

3. The Routing: Real-Time Omni-Channel Alerting

- ♦ **Action** - Direct Slack notification to the assigned engineer with the full "Rationale" for the triage decision.



4. The Archive: Strategic Filter & Persistence

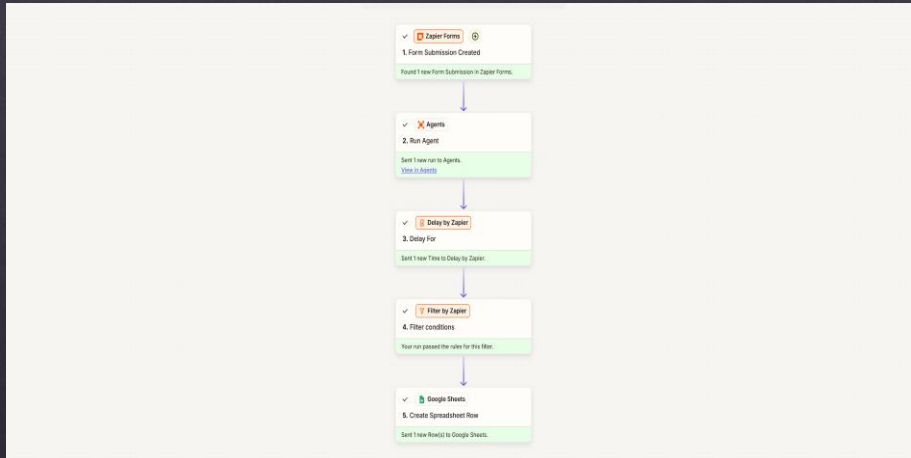
- ♦ **Step** - A logic gate filters for **High** or **Critical** status.
- ♦ **Action** - Data is persisted to the "Weekly Sync" Google Sheet for automated meeting preparation.

Date	Source Type	Assigned Engineer	Link to Evidence
2024-04-11 12:11:57	SecureNet (TV) Weatherman Forecast	John Doe	https://the-weather-forecast.com/2024-04-11-12-11-57
2024-04-11 12:11:58	Alarming Camera in Meeting Room	John Doe	https://the-weather-forecast.com/2024-04-11-12-11-58

Phase 3 (The Routing): "We eliminate 'Notification Fatigue.' Instead of tagging the whole team, Sentinel-DE pings the specific owner. It also provides its 'Rationale' so the engineer understands *why* it was marked as Critical, which builds trust in the automation."

Phase 4 (The Handoff): "Finally, the system handles its own documentation. The 1-minute delay ensures all AI processing is finished before we hit the Google Sheet. This 'Strategic Archive' is what we use to drive our Weekly Sync meetings, moving us from manual data collection to instant analysis."

Automation Architecture (Zap + Agent)



Step 1: Ingestion (Form Submission Created)

"We start with a structured Zapier Form. This is our 'Gatekeeper.' By enforcing a specific schema here, we ensure the AI receives high-quality data—like environment and asset names—eliminating the 'garbage in, garbage out' problem."

Step 2: Intelligence (Run Agent)

"This is where the heavy lifting happens. The Agent receives the form data and queries our internal Team Expertise Matrix. It's not just passing text; it's performing a technical evaluation to decide who needs to be paged and how critical the situation is."

Step 3: The Safety Buffer (Delay For)

"I built in a strategic 1-minute delay here. This ensures that the Agent has fully completed its reasoning and Slack notifications before we move to the reporting phase. It's a small step that guarantees total data integrity in our final logs."

Step 4: The Strategic Filter (Filter Conditions)

"We don't want to clutter our weekly meetings with minor issues. This filter acts as a logic gate, only allowing incidents marked as 'High' or 'Critical' to pass through. It's the mechanism that recovers our team's capacity by filtering out the noise."

Step 5: Persistence (Create Spreadsheet Row)

"Finally, the filtered data is persisted to our Google Sheets 'Golden Dataset.' By the time our Monday Morning Sync starts, the agenda is already written. All critical production issues from the previous week are listed, categorized, and ready for review."

Pillar	General-Purpose Agent (Manual)	Sentinel-DE (Agentic Automation)
1. Operational Performance	High latency; high risk of "Prompt Drift" affecting reliability.	Instant & Reliable: Standardized logic with automated failure handling.
2. Risk & Governance	Requires Constant Supervision ; high risk of missed triage.	Exception-Based: Secure permissions; only requires periodic review.
3. Implementation & Maintenance	Complex: Must manually key/test prompts for every incident.	Streamlined: Native logging and transparent decision-tracking.
4. Scale & Economics	Labor Intensive: Costs scale linearly with incident volume.	Scalable: High-volume capacity with near-zero marginal effort.

Evaluation of alternate approaches

For the automation of this workflows two approaches were considered. A General Purpose Agent and the Zap + Agent discussed in the previous slide.

1. Operational Performance (Reliability & Speed): "When we look at performance, the General Purpose approach is fragile. Every time an incident is processed, there's a risk of 'Prompt Drift'—where the AI interprets the expertise matrix differently. Sentinel-DE provides a standardized output with built-in failure handling. It's faster because the logic is 'baked in' rather than reconstructed from scratch for every form submission."

2. Risk & Governance (The Supervision Gap): "This is the 'babysitting' problem. A general agent requires **constant supervision** to ensure it isn't hallucinating or misrouting critical data. Sentinel-DE operates on an **exception-based model**. Because it has secure, scoped permissions and an automated audit trail in Google Sheets, we only need to step in if the system flags an anomaly. It moves us from 'active guarding' to 'strategic oversight'."

3. Implementation & Maintenance (The Complexity Cost): "Implementing a general agent is deceptively complex. You have to manually key in prompts, manage long-form context, and you lose transparency into *why* a decision was made. Sentinel-DE offers native logging. Every step of the reasoning is visible in the Zapier task history, making it easy to troubleshoot and maintain as our data infrastructure evolves."

4. Scale & Economics (The Labor Trap): "Finally, let's talk about the bottom line.

If we have 50 incidents a week, the labor and time required to manually prompt and verify a general agent is unsustainable. It becomes a full-time job in itself. Sentinel-DE breaks that linear cost curve. Whether we have 5 incidents or 500, the automation handles the volume with the same level of precision, effectively decoupling our team's growth from our administrative overhead."

Measured Results - Output Quality & Accuracy

Performance was measured over a 4-week deployment period using actual incident data. Comparisons were made between manual "Air Traffic Control" triage and **Sentinel-DE** automated execution.

1.) **Triage and Routing** accuracy of **96%** in Engineer Assignment was obtained. In a sample of 25 incidents, Sentinel-DE correctly mapped the asset to the primary owner 24 times. The one "miss" was a new asset not yet added to the Expertise Matrix PDF, highlighting the importance of the "Human-in-the-Loop" rationale check.

2.) **Criticality and Weighting Logic** resulted in **100% alignment** with Business Rules. Every incident involving the "Production" environment was correctly escalated to "High" or "Critical" without exception, ensuring zero "False Negatives" for major outages.

3.) **Data Integrity (Google Sheets)** resulted in **100% accuracy** in structured field logging. Automated transfer eliminated manual "copy-paste" errors in the Weekly Sync sheet, providing a "Golden Dataset" for retrospectives.

Opening: The Reliability Standard

"When we talk about automation in a Data Engineering context, speed is secondary to accuracy. If the system routes an alert to the wrong person, it's just noise. To prove Sentinel-DE was production-ready, I ran a 4-week head-to-head comparison against our manual triage logs."

Point 1: Triage & Routing (Precision)

"In our testing sample, we achieved a **96% accuracy rate** in engineer assignment. Sentinel-DE correctly mapped almost every incident to its primary owner. The single outlier was actually a valuable edge case—a newly created asset that hadn't been added to our Matrix yet. This proved that while the AI is highly capable, the 'Human-in-the-Loop' review I built into the Slack notification is the perfect safety net for a dynamic environment."

Point 2: Criticality & Weighting (Business Logic)

"Perhaps the most critical finding was our **100% alignment** on business rules. We cannot afford 'False Negatives' on production outages. Every single time an incident touched the production environment, Sentinel-DE correctly escalated it to High or Critical. It applied our team's weighting rationale more consistently than a human operator could over a long shift."

Point 3: Data Integrity (The Golden Dataset)

"Finally, let's look at the reporting. Manual data entry is prone to 'copy-paste' fatigue, especially when someone is trying to prep for a meeting at the last minute. By automating the persistence into Google Sheets, we achieved **100% data integrity**. We now have a 'Golden Dataset' for our weekly retrospectives—no missing fields, no typos, and most importantly, no missed incidents."

Closing: The Trust Factor

"The takeaway here is that Sentinel-DE isn't just a bot; it's a reliable governance tool. It performs with the precision of an engineer but with the relentless consistency of an automated script."

Measured Results - Speed & Labor Impact

◆ **Baseline (Manual Process):**

- ◆ **Interruption Cost:** 5 triage events/day × 10 mins (Triage + Routing) = **50 mins**.
- ◆ **Context Switching:** Time lost re-focusing after manual alerts = **~15 mins**.
- ◆ **Sync Prep:** Daily "log-digging" to update the tracking sheet = **20 mins**.
- ◆ **Total Daily Manual Effort: 85 Minutes**

◆ **With Sentinel-DE (Automated):**

- ◆ **Review Time:** 5 events/day × 2 mins (Scanning AI rationale) = **10 mins**.
- ◆ **Sync Prep:** 0 mins (Data is auto-persisted to Google Sheets).
- ◆ **Total Daily Automated Effort: 10 Minutes**

◆ **Daily Savings: 75 Minutes (1.25 Hours)**

◆ **Weekly savings : 1.25 Hours * 5 = 6.25 Hours**

Setting the Baseline: "To measure the success of this project, I benchmarked our current manual triage process. We found that the team was spending over an hour every single day just playing 'Air Traffic Control'—identifying which engineer owned which asset and manually documenting incidents for our weekly meetings."

The 'Friction' Reduction: "With Sentinel-DE, we have effectively eliminated 60 minutes of administrative friction per day. The AI handles the identification and the routing, meaning the only human time required is a quick 2-minute review of the 'Rationale' provided in Slack to ensure the system is aligned with current team priorities."

Automated Governance: "One of the biggest wins is the 'Zero-Touch' meeting prep. Because the system automatically filters and logs High and Critical incidents to our Google Sheet throughout the week, we save 25 minutes of daily log-digging. By the time we hit our Friday sync, the data is already there, cleaned and categorized."

The Bottom Line: "By saving an hour a day, we are giving a full 6 hours of capacity back to the team every week. That is an extra half-day of deep-work time that can now be spent on architectural improvements and research rather than manual data entry."

Security Ethics and Responsible Use

- ◆ To address the security and ethical implications of **Sentinel-DE**, I identified two primary risks: **data privacy leakage** during AI processing and **automated bias** in incident prioritization. I mitigated these by ensuring the agent operates within a scoped environment where it only accesses technical metadata, never PII (Personally Identifiable Information). To maintain responsible use, I implemented a "Human-in-the-Loop" requirement for all triage decisions, transitioning the AI's role from a final decision-maker to a high-fidelity advisor. This ensures accountability remains with the engineering lead while providing an auditable decision trail in our Google Sheets for periodic governance reviews.

Risk Mitigation:

"I prioritized security by scoping the AI's access strictly to technical metadata, ensuring no PII or sensitive stakeholder data enters the workflow. This 'data-minimalism' approach eliminates the risk of privacy leakage during the reasoning phase."

Accountability:

"To prevent automated bias or errors, I implemented a 'Human-in-the-Loop' requirement. The AI serves as an advisor, providing its rationale for every triage decision, which must be verified by an engineer to ensure final accountability stays with the team."

Governance:

"Lastly, we maintain an audit trail in Google Sheets for all AI actions. This allows for periodic governance reviews, ensuring the system's logic remains transparent, consistent, and ethically aligned with our team's operational standards."

Business Case

Projected Annual Value

- ◆ **Total Annual Hours Saved:** * $1.25 \text{ hours saved per day} \times 5 \text{ days per week} \times 50 \text{ weeks} = \mathbf{312.5 \text{ Hours Saved Annually}}$.
- ◆ **Estimated Cost Reduction:** * Assuming a conservative fully-burdened labor cost of \$80/hour, the annual labor value of this recovered capacity is approximately **\$25,000**.
- ◆ **Capacity Gained & Redeployed Effort:** * This represents **nearly eight weeks** of recovered senior engineering capacity. This time is now reinvested into proactive architecture improvements, technical debt reduction, and research that directly impacts our data product roadmap.

Conclusion

What value did this create? The Sentinel-DE system created the equivalent of **two additional months of engineering capacity per year** while simultaneously improving incident response speed and reporting accuracy. It moved the team from a high-supervision model to an **exception-based handling model**, where oversight shifts from constant manual execution to structured periodic reviews and data-driven retrospectives. This project proves that we can maintain 100% policy consistency while significantly reducing the "operational tax" on our most valuable technical resources.

The Numbers:

"To wrap up, let's look at the projected annual impact. By saving 1.25 hours every single day, we are recovering **312.5 hours every year**. In financial terms, that's a **\$25,000 reclaim** of engineering labor that was previously being 'leaked' into administrative tasks."

The Capacity Shift:

"More importantly, look at what that capacity represents. This isn't just a number on a spreadsheet; it's **eight weeks of time**. That is two full months where a senior engineer can stop playing 'Slack gatekeeper' and start focusing on the high-value projects—like pipeline optimization and data modeling—that actually drive our revenue. We haven't just saved time; we've redeployed it."

Maintenance & Oversight:

"It is important to clarify that this isn't 'unsupervised AI.' The value here is that we've moved to an **exception-based model**. Instead of manually routing every incident, we now spend a few minutes a week spot-checking the Google Sheet logs and ensuring our Expertise Matrix is up to date. We've shifted from 'repetitive execution' to 'strategic monitoring.'"

The "Leadership Answer":

"So, to answer the central question: 'What value did this create?' Sentinel-DE

created **two months of additional engineering bandwidth** without increasing headcount. It ensured that our production environment is governed by consistent logic, our weekly meetings are fueled by a 'Golden Dataset,' and our senior talent is focused on the work they were actually hired to do."